To add (+5) and (–3), We start at zero facing positive direction and move five steps forward to represent (+5). Since the operation is addition we maintain the same direction and move three units backward to represent (–3). We land at +2. So, (+5)+(–3)=2. This is shown in Fig.1.5.

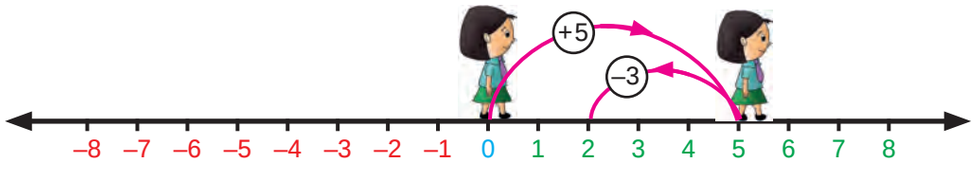


Fig. 1.5

Proceeding in the same way let us try another example. Find the sum of (–6) and (–4). We start at zero facing positive direction continuing in the same direction and move 6 units backward to represent (–6) and in the same direction move 4 units backward to represent (–4). We land at (–10).

Therefore, (–6)+(–4)=–10

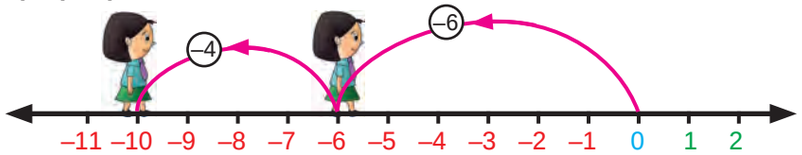


Fig. 1.6

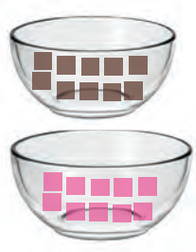
Try this

Find the value of the following using the number line activity:

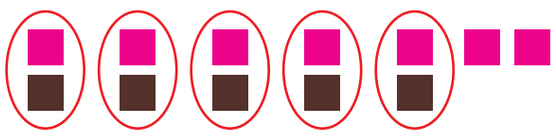
(i) (–4) + (+3) (ii) (–4) + (–3) (iii) (+4) + (–3)

Activity

There are two bowls with tokens of two different colours brown and pink. Let us denote one brown token by (+1) and one pink token by (–1). A pair of tokens, brown (+1) and pink (–1), will denote zero [1+(–1)=0]



To add two integers, we pick the required number of tokens and form possible zero pairs. The remaining number of tokens left after pairing is the sum of the two integers.



(–7)
(+5)
–2

To add (–7) and (+5) we pick 7 pink tokens and 5 brown tokens. We form zero pairs from the tokens as above. We can form 5 zero pairs. Then we are left with 2 pink tokens. Hence, $(-7)+(+5)=-2$.

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